



2026 Level 2 - Corporate Issuers

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Analysis of Dividends and Share Repurchases

- a. describe the expected effect of regular cash dividends, extra dividends, liquidating dividends, stock dividends, stock splits, and reverse stock splits on shareholders' wealth and a company's financial ratios
- b. compare theories of dividend policy and explain implications of each for share value given a description of a corporate dividend action
- c. describe types of information (signals) that dividend initiations, increases, decreases, and omissions may convey
- d. explain how agency costs may affect a company's payout policy
- e. explain factors that affect dividend policy in practice
- f. calculate and interpret the effective tax rate on a given currency unit of corporate earnings under double taxation, dividend imputation, and split-rate tax systems
- g. compare stable dividend with constant dividend payout ratio, and calculate the dividend under each policy
- h. compare share repurchase methods
- i. calculate and compare the effect of a share repurchase on earnings per share when 1) the repurchase is financed with the company's surplus cash and 2) the company uses debt to finance the repurchase
- j. calculate the effect of a share repurchase on book value per share
- k. explain the choice between paying cash dividends and repurchasing shares
- l. describe broad trends in corporate payout policies
- m. calculate and interpret dividend coverage ratios based on 1) net income and 2) free cash flow
- n. identify characteristics of companies that may not be able to sustain their cash dividend

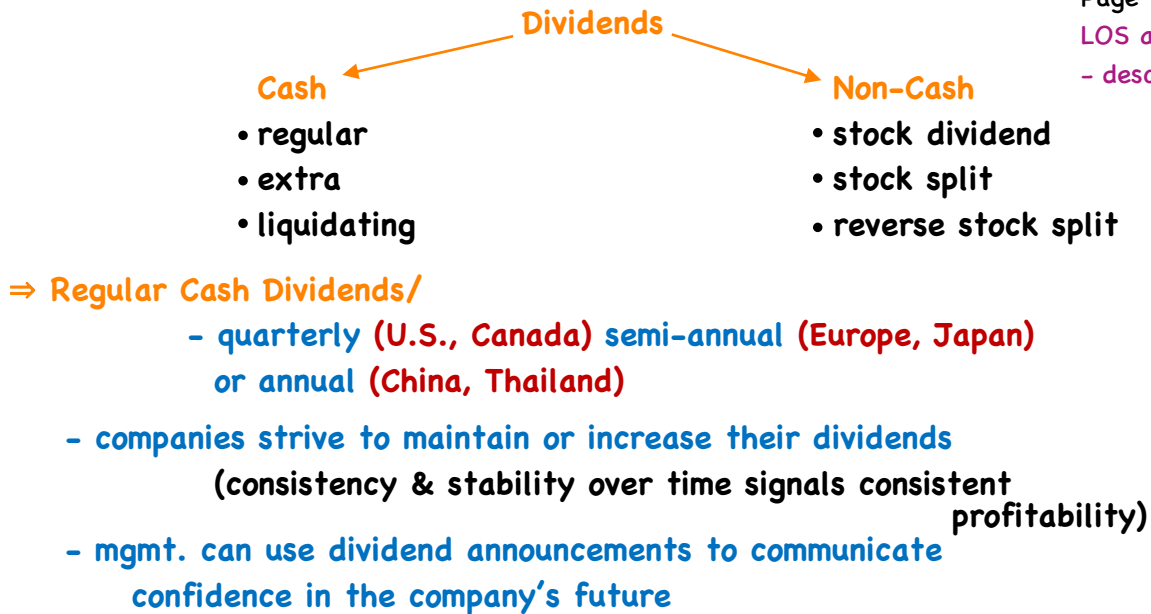
LOSs will match between the video and the PDFs, but may be in a different order than the CFAI readings

Cash vs. Non-Cash Dividends

Page 1

LOS a

- describe



Page 2

LOS a

- describe

- ⇒ Regular Cash Dividends/
- DRiPs – Dividend Reinvestment Plans – rather than cash dividends, more shares are paid out
 - ① open-market DRP – company purchases shares in the open market to deliver
 - ② new-issue DRP – shares to deliver are issued
 - ③ hybrid – either ① or ②
- Company benefits/
- may encourage a more diverse shareholder base (smaller shareholders have an easy means to accumulate shares)
 - may stimulate longer-term investment
 - new-issue DRPs allow the company to raise new equity capital without floatation costs

⇒ **Regular Cash Dividends/**

- DRiPs – Dividend Reinvestment Plans

Shareholder benefits/

- allow the accumulation of shares using cost averaging
- typically no transaction costs
- new-issue DRPs (but not exclusively) may allow shares at 2%-5% market discount

Shareholder disadvantages/

- extra record keeping (ACB)
- still taxable

⇒ **Extra (Special) Dividends/** a dividend paid by a company that does not pay a div. on a regular schedule or a div. in addition to a regular cash div.

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- describe

⇒ **Extra (Special) Dividends/**

- a means of distributing more earnings only during strong earnings years

⇒ **Liquidating Dividends/** - used when a company

- 1/ goes out of business & the net assets are distributed to shareholders
 - 2/ sells a portion of its business for cash and the proceeds are distributed to shareholders
 - 3/ pays a dividend that exceeds its accum. Ret. Earnings
- is a return of capital rather than a distribution from earnings

⇒ **Stock Dividends/** - company distributes additional shares
- sh. total cost basis remains the same, cost/sh. is reduced

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LOS a

- describe

⇒ Stock Dividends/

- generally not taxable to shareholders
- will not affect sh. proportional ownership
- also will not affect the value of each sh. ownership position

	Before Dividend	After Dividend
Shares outstanding	1,000,000	1,030,000
Earnings per share	\$1.00	\$0.97 (1,000,000/1,030,000)
Stock price	\$20.00	\$19.4175 (20 × 0.9709)
P/E	20	20
Total market value	\$20 million	\$20 million (1,030,000 × \$19.4175)
Shares owned	100,000 (10% × 1,000,000)	103,000 (10% × 1,030,000)
Ownership value	\$2,000,000 (100,000 × \$20)	\$2,000,000 (103,000 × \$19.4175)

- Adv./
- favours long-term investors which may lower the company's cost of equity financing (basically issuing shares = cash div.)
 - helps increase the stock's float, improves liquidity, dampens price volatility

⇒ Stock Dividends/

- more likely to keep a share price within an optimal range
- Cash Div. affect Capital Structure, no so with stock Div.
 - reduce assets
 - reduce Sh. Equity
 - liquidity ratios decrease
 - financial leverage ratios increase (D/A, D/E)
- do not affect assets
- do not affect Sh. Equity
- Ret. Earn. \$↓
- Cont. Cap. \$↑
- no affect on liquidity or leverage ratios

⇒ Stock Splits/ 2-for-1 ① price/2 ② shares × 2

EPS/2 P/E - unch. MV_e - unch. Div yield - unch.

- no affect on any shareholder accounts

⇒ Stock Splits/

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- describe

	Before Split	After Split
Number of shares outstanding	4 million	8 million
Stock price	€40.00	€20.00 (€40/2)
Earnings per share	€1.50	€0.75 (€1.50/2)
Dividends per share	€0.50	€0.25 (€0.50/2)
Dividend payout ratio	1/3	1/3
Dividend yield	1.25%	1.25% (€0.25/€20.00)
P/E	26.7	26.7 (€20.00/€0.75)
Market value of equity	€160 million	€160 million (€20.00 × 8 million)

⇒ Reverse Stock Splits/

- increases the share price and reduces the # of outstanding shares
- no affect on MV_e or sh. total cost basis

Dividend Policy Theories

⇒ Does a dividend payment increase or decrease the value of a firm to its owners?

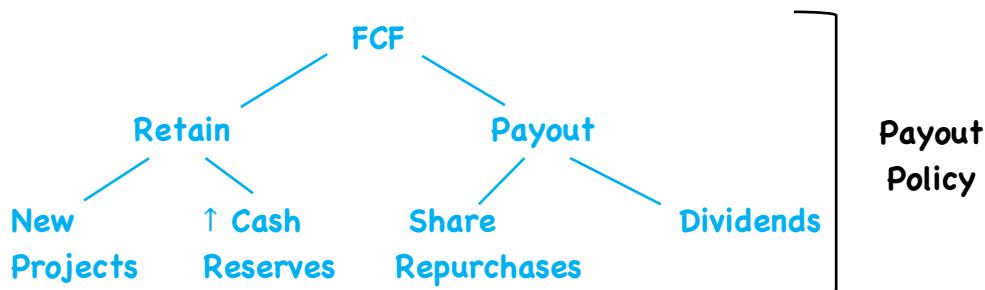
- if so, what should a firm's payout policy be?

Page 8

LOS b

- compare

- explain



- to answer these questions, first start by making the payout policy irrelevant

Page 9

LOS b

- compare
- explain

1) Dividend Irrelevance Theory/

- Miller, Modigliani (1961)

- assume perfect capital markets

- 1) no taxes
- 2) no transaction costs
- 3) no agency costs
- 4) information symmetry

- div. policy will have no effect on CoC or shareholder wealth

⇒ Investors can create their own dividend

∴ company div. policy adds no value

- div. & sh. repurchases are interchangeable
- value of firm depends only on net income produced by assets, not on how this income is split (Div. vs. RE)

need
1 & 2
above

Page 10

LOS b

- compare
- explain

2) Dividend Preference Theory

- Gordon (63), Lintner (62)

⇒ stocks risk declines as

dividends increase i.e. $r_e \downarrow$, WACC $\downarrow$

- dividends are a sure thing, retained earnings are risky
'a bird in the hand' argument

∴ shareholders prefer dividends and are willing to accept a lower required return on equity

- also reduces agency costs (reduces the risk that mgmt. will misuse the cash)

3) Tax Effect Theory

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LOS b

- compare
- explain

- if $T_D > T_{CG}$, investors prefer low payout ratios or repurchases as a form of distribution
 - investors would be willing to pay more for low payout companies
- if $T_D < T_{CG}$ - investors prefer cash dividends
- if $T_D = T_{CG}$ - investors still prefer capital gains to dividends (time value of money)

Dividend Signals

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- MM $\Rightarrow$ information symmetry
- in reality, asymmetry between mgmt. & investors
 - $\therefore$ may be some info in dividend policy changes
- dividend initiations/increases $\Rightarrow$ signal that mgmt. expects higher earnings in the future
 - mgmt. may be motivated to initiate/increase if they believe shares to be undervalued
- dividend omissions/decrease/failures to increase - convey negative information

Clientele/Agency Issues

⇒ Agency Costs/

- Jensen's FCF hypothesis/ by paying out a high ratio of earnings, mgmt. is constrained from overinvestment
- if the company is leveraged, high payout ratio may constrain available cash leading to underinvestment
 - high payout ratio increases agency costs to creditors
 - typically have debt covenants restricting the level/amount of dividends

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- explain

Factors Affecting Dividend Policy

1) Investment opportunities/

- a company with many inv. opportunities will tend to payout less in dividends
- internally generated cash flows are a cheaper source of equity funding

2) Expected volatility of future earnings/

- the more volatile earnings are, the greater the risk that a dividend increase may not be met by future earnings

3) Financial flexibility/

- may use share repurchases rather than initiate/increase a dividend

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LOS e

- explain

4) Tax Considerations/

- Taxation Methods

- double taxation system
 - earnings taxed at corporate level, then again at individual level
- dividend imputation tax system
 - corporate profits are taxed just once, at the shareholder tax rate
- split-rate tax system
 - corporate profits destined as dividends are taxed lower than profits to be retained

5) Floatation Costs/

- fees a company must pay to sell new shares plus the negative market price reaction
- higher (as a %'age of funds raised) for smaller firms
 - ∴ try to avoid a level of dividends that would create the need to raise new equity to finance NPV⁺ projects

6) Contractual/Legal Restrictions

- some countries ⇒ companies must pay a div.
- other restrict payment (i.e. impairment of capital rule)
- bond covenants as well
- cumulative divs. on preferred shares before common are paid

Taxation Systems

1) Double Taxation System/

	39.6%	15.0%
Net income before taxes	US\$100	US\$100
Corporate tax rate	35%	35%
Net income after tax	US\$65	US\$65
Dividend assuming 100% payout	US\$65	US\$65
Shareholder tax on dividend	US\$25.74	US\$9.75
Net dividend to shareholder	US\$39.26	US\$55.25
Double tax rate on dividend distributions*	60.7%	44.8%

- different
personal tax
rates

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LOS f

- calculate
- interpret

2) Dividend imputation tax system/

	Marginal Shareholder Tax Rate	
	15%	47%
Pretax income	A\$100	A\$100
Taxes at 30% corporate tax rate	30	30
Net income after tax	70	70
Dividend assuming 100% payout	70	70
Shareholder tax on pretax income	15	47
Less tax credit for corporate payment	- 30	- 30
Tax due from shareholder	(15)	17
Effective tax rate on dividend	15/100 = 15%	47/100 = 47%

- one higher, one lower
than corporate rate

- tax credits can be used
to offset taxes owed
on other income

3) Split-rate tax system

Pretax earnings	€200
Pretax earnings retained	100
35% tax on retained earnings	35
Pretax earnings allocated to dividends	100
20% tax on earnings allocated to dividends	20
Dividends distributed	80
Shareholder tax rate	35%
After tax dividend to shareholder	$[(1 - 0.35) \times 80] = 52$
Effective tax rate on dividend	$[20\% + (80 \times 0.35)\%] = 48\%^*$

loss of \$48
= 48% tax

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LOS f

- calculate
- interpret

Dividends/Share Repurchases

⇒ Dividend Policy/

1) Stable Dividend Policy/ - most common

- dividends based on long-term forecast of sustainable earnings

$$\underbrace{E(\text{increase in Div.}) = \left[E(\text{earnings}) \times \text{target DPR} - \text{Previous Div.} \right] \times \frac{1}{\text{time frame}}}_{\text{moving towards a target DPR}}$$

moving towards a target DPR

2) Constant Dividend Payout Ratio

- div. fluctuate w/ earnings

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LOS g

- compare

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The dividend policy of Berkshire Gardens Inc. can be represented by a gradual adjustment to a target dividend payout ratio. Last year Berkshire had earnings per share of US\$3.00 and paid a dividend of US\$0.60 a share. This year it estimates earnings per share will be US\$4.00. Find its dividend per share for this year if it has a 25% target payout ratio and uses a five-year period to adjust its dividend.

Screen 21 intentionally left blank

Share Repurchase Methods

- repurchase (buyback) – a transaction in which a company buys back its own shares
 - can be viewed as an alternative to cash dividends
 - repurchased shares classified as:
 - 1) treasury shares – if they may be reissued
 - 2) canceled shares – if they will be retired
 - not considered for dividends, voting, calculating EPS
- U.S. ⇒ 1980s Dividends > 5x sh. repurchase MV
late 90s/early 2000s sh. repurchases MV > Dividends
- use rising elsewhere
 - authorized repurchases are not contractual (no commitment) as declared dividend would be

⇒ **Methods/** (in order of importance)

1/ Buy in the open market (as conditions warrant)

- most common, most flexible (no legal obligation)
- no shareholder approval required in U.S., but is required in Europe
- tends to provide price support, most cost effective

2/ Buy back a fixed number of shares at a fixed price

- a fixed price tender offer (usually at a premium to the current market price (adv. → speed))

3/ Dutch auction - specify a price range, solicit bids, find a price that clears the lot (adv. → speed)

4/ Repurchase by direct negotiation - typically with a major shareholder often at a premium

- may be at a discount if seller needs liquidity

Repurchase with Cash or Debt

- **repurchases affect both the BS & IS**

- if made with cash → assets & sh. equity decline
 - leverage increases
- if financed with debt → leverage increases even more
- could increase EPS depending on 1) how repurchased financed
2) what cost/share

- cash repurchase, no change in NI ⇒ higher EPS
- debt financed repurchase, Int. Exp. lowers NI ⇒ uncertain effect

e.g./ 10 million shares

\$100 million NI

EPS = $\frac{100}{10} = \$10/\text{sh.}$

Buyback @ \$140/sh.

Cash = \$240 million

(earning almost \$0)

$$\text{shares retired} = \frac{\$240\text{M}}{\$140} = 1,714,286$$

$$\begin{aligned} \text{new EPS} &= \frac{\$100\text{M}}{(10\text{M} - 1.714286\text{M})} \\ &= \$12.07/\text{sh.} \end{aligned}$$

e.g. 2/ Borrow \$12M $\Rightarrow$ Planned repurchase = 200,000 sh.

$$P_0 = \$60$$

$$NI = \$6.6M$$

$$EPS = \$3$$

$$P/E = \$60 / 3 = 20x$$

$$E/P = \$3 / 60 = 5\%$$

$$\text{Shares} = 2.2M$$

a) $r_b(1 - t) = 5\%$

b) $r_b(1 - t) = 6\%$

a) $EPS = ? = NI / \text{share \#}$

$$\frac{6.6M - (12M \times .05)}{2.2M - 200,000} = \frac{6M}{2M} = \$3/\text{sh.}$$

- note that $E/P = 5\% = r_b(1 - t)$

b)

$$\frac{\$6.6M - (12M \times .06)}{2.2M - 200,000} = \frac{5.88M}{2M} = \$2.94/\text{sh.}$$

- note $r_b(1 - t) > E/P$

$$6\% > 5\%$$

- internal funds $\Rightarrow \uparrow$ EPS (if funds could not earn their cost of capital)
- external fund $\Rightarrow \uparrow$ EPS if $r_b(1 - t) < E/P$
 $\downarrow$ EPS if $r_b(1 - t) > E/P$

Effect on Book Value

Company A

$$P_0 = \$20 > BVPS$$

$$\text{shares} = 10M$$

$$\text{buyback} = \$5M$$

$$BV_e = \$100M$$

$$BVPS = \$100M / 10M = \$10$$

$$\text{buyback} = \$5M / \$20 = 250k$$

$$\text{new share \#} = 9.75M$$

$$BV_e = \$95M$$

$$BVPS = \$95M / 9.75M = \$9.74$$

when $P_0 > BVPS$, $BVPS \downarrow$

Company B

$$P_0 = \$20 < BVPS$$

$$\text{shares} = 10M$$

$$\text{buyback} = \$5M$$

$$BV_e = \$300M$$

$$BVPS = \$300M / 10M = \$30$$

$$\text{buyback} = \$5M / \$20 = 250k$$

$$\text{new share \#} = 9.75M$$

$$BV_e = \$295M$$

$$BVPS = \$295M / 9.75M = \$30.26$$

when $P_0 < BVPS$, $BVPS \uparrow$

Cash vs. Repurchase

1) Potential tax advantages/

- T_D vs. T_{CG}
- T_D lower $\Rightarrow$ Divs.
- T_{CG} lower $\Rightarrow$ repurchase

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LOS k

- explain

2) Share price support/

- repurchase signals that mgmt. considers shares undervalued or/
- may be seen that the company has few profitable investment opportunities

3) Added managerial flexibility

- repurchases do not have to repeat or even occur once announced

4) Offsetting dilution from employee stock options

5) Increasing financial leverage/

- repurchases modify the company's capital structure $\Rightarrow$ decrease equity $\Rightarrow$ increase leverage

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LOS k

- explain

e.g./ D:E = 30:70

Debt = \$30M

Repurchase = \$7M

New D:E if Cash ?
Debt ?

leveraged
recapitalization

$\Rightarrow$

Cash/	Debt	\$30M	-	32%
	Equity	63M	-	68%
		\$93M		
		D/E = .4706		

Debt/	Debt	\$37M	-	37%
	Equity	63M	-	63%
		100M		
		D/E = .5873		

- Sh. repurchase is effectively an alternative to paying a special cash dividend
- Sh. repurchases tend to increase in volume when the economy is strong & companies have more cash

e.g./

Income Statement		Statement of Cash Flows	
Sales	1,200	Cash flow from operations	135
Earnings before taxes	155	Cash flow from investing activities	(84)
Taxes	35	Cash flow from financing activities:	
Net income	120	Debt repayment	(4)
		Share repurchase	(32)
		Proposed dividend	(15)
		Estimated change in cash	0

- does a div. suggest that this company is no longer a growth company?

- Identify the div. policy (40¢/sh. initiation)

Ratios		Five-Year Forecasts	
Current ratio	2.1	Sales growth	8% annually
Debt/Equity (at market)	0.27	Earnings growth	11% annually
Interest coverage	10.8x	Projected cost of capital	10%
ROA	10.0%		
ROE	19.3%		
P/E	20x		
E/P	5.0%		

Trends in Dividend Policies

- catering theory (Baker & Wurgler, 2004)
 - companies adapt their dividend policy over time to changing investor tastes
- %'age of index components in U.S. paying divs. lower than comparable European index components
- fraction of companies paying cash dividends has been in long-term decline in most developed markets
 - however, DPRs have increased over-time
- since early 1980's in U.S., and early 1990s in UK and Europe, fraction of companies using share repurchases has trended upwards

Dividend Coverage Ratios

A/ $DPR = \text{Div.} / NI$

$\text{Dividend Coverage Ratio} = 1 / DPR = NI / \text{Div.}$

- higher DPR, lower DCR $\Rightarrow$ greater risk of Div. cut

- generalizations/

• large mature companies

DPR $\sim$ 40 – 60%

DCR $\sim$ 1.75x – 2.5x

• if DCR < 1 , div. may be in jeopardy, unless an event was to blame for the drop in earnings

• fast growing industries/new companies – no or low DPRs

B/ $\text{FCFE Coverage Ratio} = \frac{\text{FCFE}}{\text{Div.} + \text{Repurch.}}$

CFO – FCInv – WCInv

= 1 - all cash returned to shareholders

> 1 - improving liquidity

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LOS m

- calculate

- interpret

Dividend Safety

$\Rightarrow$ unsustainable policies/

$\Rightarrow DPR > 100\%$

- either using NI or FCFE

$\Rightarrow DCR < 1$

$\Rightarrow \text{FCFE} / (\text{Div.} + \text{Repurch.}) < 1$

$\Rightarrow$ extremely high Div. yields versus existing bond-yields

(D/P_0)

$\Rightarrow$ borrowing to pay dividends

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LOS n

- identify

Environmental, Social, and Governance (ESG) Considerations
in Investment Analysis

- a. describe global variations in ownership structures and the possible effects of these variations on corporate governance policies and practices
- b. evaluate the effectiveness of a company's corporate governance policies and practices
- c. describe how ESG-related risk exposures and investment opportunities may be identified and evaluated
- d. evaluate ESG risk exposures and investment opportunities related to a company

Corporate Governance/ESG

Page 1

LOS a

- describe

• ownership structures/

→ Dispersed – many shareholders, none which has the ability, individually, to exercise control

→ Concentrated – an individual shareholder (or group) with the ability to exercise control

more common globally

→ family
→ another company
→ sovereign entity

Note: controlling shareholders may even be minority shareholders

due to: a) horizontal ownership

< 50% ownership

- key customers/suppliers with cross-holdings

or/ vertical ownership → company → holding company → operating company

b) dual-class shares – one class represents a minority ownership interest with a majority of voting rights

Page 2

LOS a

- describe

• ownership structures/

1/ Dispersed ownership, dispersed voting power

• weak shareholders, strong managers

principal-agent conflicts

- can be mitigated if controlling shareholders are present

2/ Concentrated ownership, concentrated voting power

• strong shareholders, weak managers

can allocate company resources to their benefit (with a minus sign) can monitor mgmt. (with a plus sign)

principal-principal problem

majority

minority

• ownership structures/

3/ Dispersed ownership, concentrated voting power - dual-class

principal-principal conflict

weak majority

strong minority

4/ Concentrated ownership, weak voting power

- result of restrictions on voting rights of large share positions → voting caps

- types of shareholders that have an influence on corporate governance

1/ Banks (notably in Europe/Asia) - both lender and shareholder

- potential conflict of interest vs. other shareholders

2/ Families (predominant from in Latin America, common in Asia/Europe)

+/ • lower risks associated with principal-agent problems
(concentrated ownership with managers)

-/ • poor transparency, lack of mgmt. accountability, minimal consideration for minority shareholder rights, difficulty in attracting talent

3/ State-owned enterprises (SOEs) - listed SOEs are owned by the sovereign gov't. but also have shares traded on a public exchange (mixed-ownership model)

Page 3c

LOS a

- describe

3/ State-owned enterprises (SOEs) - tend to have lower scrutiny of mgmt. (implicit/explicit gov't. guarantee)

- may pursue social/public policy considerations at the expense of maximizing shareholder value

4/ Institutional Investors (MFs, Pension funds, Insurance Cos., HFs)

- collectively represent a significant proportion of equity market ownership
- significant power to hold mgmt./BoD accountable

5/ Group Companies - may engage in related-party transactions at the expense of minority shareholders

6/ Private Equity firms - strategic investors → presence tends to promote corporate governance development

Page 3d

LOS a

- describe

7/ Foreign Investors

- when foreign ownership > domestic ownership (from countries with more stringent standards)
 - promotes corp. gov. development
 - local minority shareholders benefit
 - especially in emerging markets

8/ Managers and Board Directors (insiders)

- as insider ownership ↑, more likely to adopt a long-term focus
 - economic interests aligned with external shareholders
- However, too much ownership may give rise to principal-principal conflicts

Page 4

LOS a

- describe

• Effects on Corporate Governance/

1/ Director Independence - independent director - no material employment, ownership, remuneration

- Independence higher with dispersed ownership structures
- helps mitigate the principal-agent conflict

2/ Board Structures → one-tier: single board w/ internal & external directors (most common)

→ two-tier: supervisory board that oversees a management board

- supervisory board mitigates strong manager

3/ Special Voting Arrangements - mitigates concentrated voting power

- independent directors must be approved by
 - 1/ majority vote
 - 2/ majority vote of minority shareholders

Page 5

LOS a

- describe

• Effects on Corporate Governance/

4/ Corporate Governance Codes/Laws/Listing Requirements

- many countries require adoption of governance codes
- comply or explain

5/ Stewardship Codes - voluntary codes → encourage investors to exercise their legal rights and increase their level of engagement

- for some countries, institutional investors must 'comply or explain'

1/ Board policies/practices

a) Structure - CEO/Chair separation?

↓
should be independent

b) Board Independence - %'age or number?

c) Board Committees - Audit, Governance, Remuneration, Nomination, Risk/Compliance - Independent members?

d) Board skills/experience - appropriate and diverse given the nature of the business

- tenure: > 10 yrs. = long - ⊕ comprehensive understanding of the business
/

⊖ may affect independence

e) Board Composition - diversity (age, gender, tenure)

- number (smaller boards may govern more effectively)

f) Other Considerations - Board evaluation

(self or external review)

2/ Executive Remuneration

- transparency of compensation
- performance criteria for incentive plans
- link with company strategy
- pay differential vs. average worker

appropriate or excessive?

} clawback if certain events occur

- misconduct
- breach of law etc...

3/ Shareholder voting rights

- Straight voting - one vote for each share
- dual-class → more than 1 vote/share

- Identifying ESG-related risks

- what is material will depend on the investment horizon

- relevant factors: E ?
S ?
G ?
company/
industry

- identifying a company's/industry's ESG factors

1/ proprietary methods → analysts use their own judgment or firm's methods

2/ ESG data providers (MSCI, Sustainalytics)

3/ Not-for-profit industry organizations or initiatives

- provide data & insight on ESG issues

e.g. International Integrated Reporting Council

Global Reporting Initiative

Equity vs. Fixed Income → generally only focused on downside risk

↳ opportunities & downside risks

- analyzing ESG-related factors:

Equity • forecasting financial metrics (op. Inc.) or ratios
• adjusting valuation model variables (discount rate)
• sensitivity or scenario analysis

Credit analysis • affects on risk (cash flow impacts, credit spreads)
- may differ by maturity

- ESG disclosures range from minimal to comprehensive data → creates potential comparability issues for analysts **Ex. #2**

• Global Reporting Initiative (globalreporting.org)

- develops sustainability reporting standards
- includes a list of business activity groups
with relevant sustainability topics that correspond to each group

- Sustainable Accounting Standards Board (sasb.org)

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LOS c

- describe

- promote uniform accounting standards for sustainability reporting
- developed the SASB Materiality Map
 - lists relevant ESG-related, sector-specific factors that are deemed material Ex. 4

- Green Bonds → proceeds designated to fund specific projects that have environmental or climate benefits
 - issuers typically incur additional costs related to the monitoring and reporting of the use of the bond's proceeds
 - But// may sell at a premium if demand is strong (tighter spreads)
- typically resemble an issuer's conventional bonds, with same credit rating and bondholder recourse

- Greenwashing - risk that the bond's proceeds are not actually used for a beneficial environmental or climate-related project

Page 11

LOS c

- describe

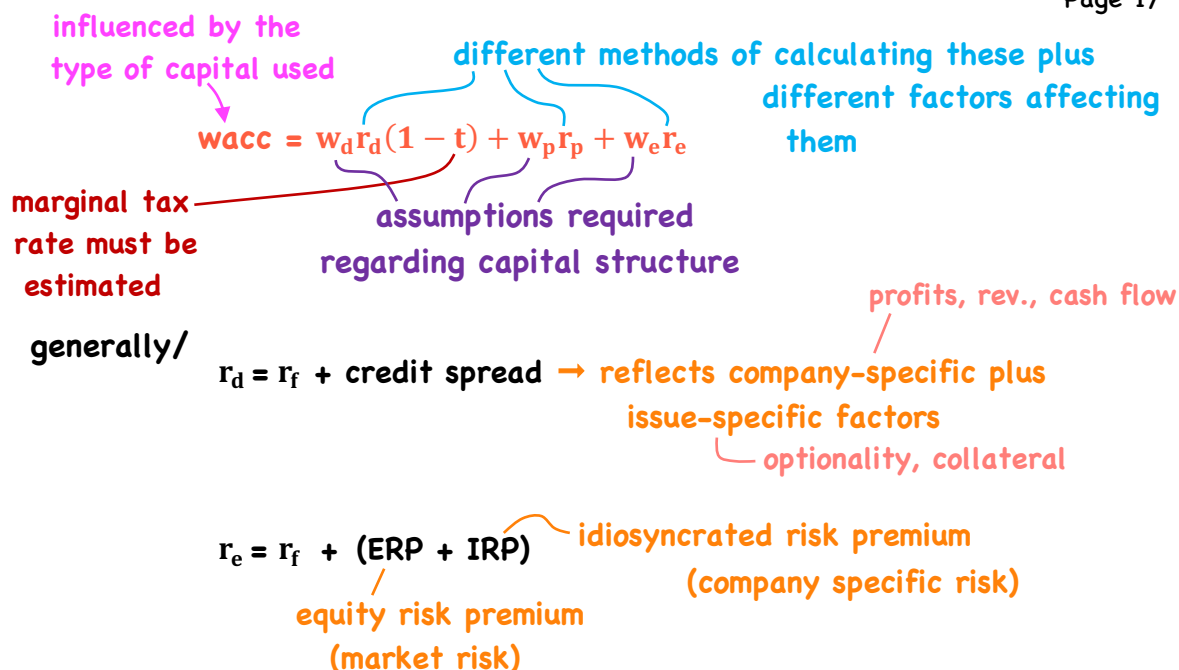
- Liquidity → typically purchased by buy-and-hold investors

Cost of Capital: Advanced Topics

- a. explain top-down and bottom-up factors that impact the cost of capital
- b. compare methods used to estimate the cost of debt.
- c. explain historical and forward-looking approaches to estimating an equity risk premium
- d. compare methods used to estimate the required return on equity
- e. estimate the cost of debt or required return on equity for a public company and a private company
- f. evaluate a company's capital structure and cost of capital relative to peers

Cost of Capital

Page 1/



* Explain top-down and bottom-up factors that impact the cost of capital

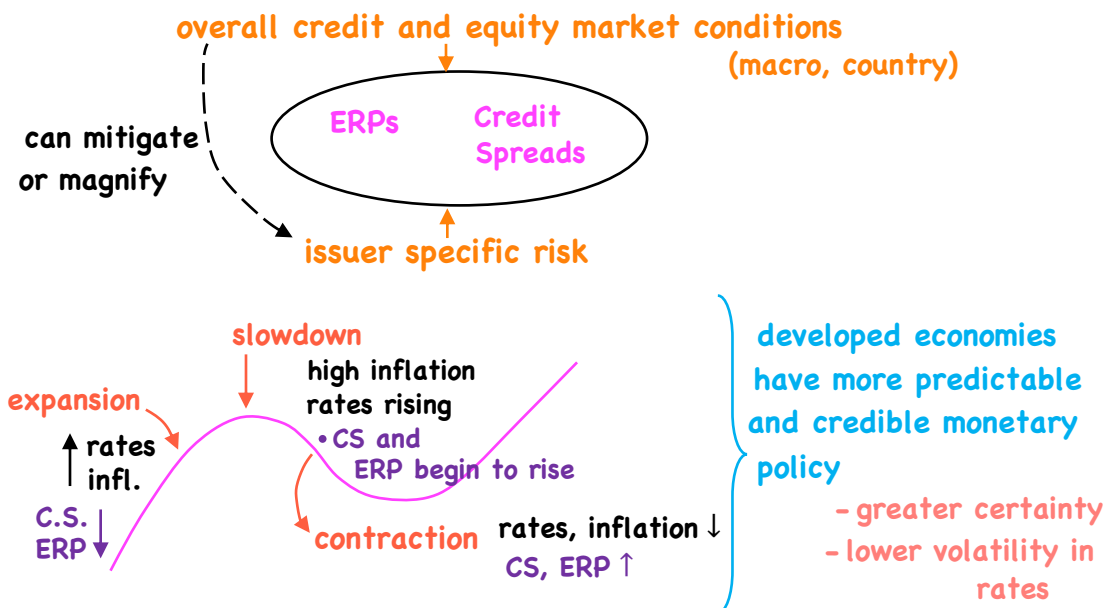
Page 2/

• Cost of Capital Factors



* Explain top-down and bottom-up factors that impact the cost of capital

2/ Market conditions e.g. interest rates, inflation, macro-environment



* Explain top-down and bottom-up factors that impact the cost of capital

3/ Legal/Regulatory considerations, Country risk

- common law systems - greater enforceability of investor rights

∴ more developed capital markets vs. civil law systems

- combined with more mature regulatory environments

= lower CS, ERPs

4/ Tax Jurisdiction - the higher the marginal tax rate, the greater the use of debt in the capital structure

• Bottom-up factors/

	① Revenue, Earnings, Cash Flow Volatility	Effect on Cost of Capital	
		Lower	Higher
recurring or subscription-based customer concentration - majority of revenue from a few customers	Higher stability of revenues, earnings, and cash flows	✓	
	Higher revenue concentration		✓
	Higher earnings predictability	✓	
	Higher operating leverage		✓
	Higher financial leverage		✓
	Higher ESG risks		✓

greater earning volatility

higher vol. = higher risk

* Explain top-down and bottom-up factors that impact the cost of capital

② Asset nature and liquidity

Asset Type	Effect on Cost of Capital	
	Lower	Higher

Higher proportion of fungible, tangible assets

✓

→ can be pledged as collateral

Higher proportion of liquid assets

✓

- lower cost of debt but may raise the cost of equity

③

Financial Strength	Effect on Cost of Capital	
	Lower	Higher

Higher profitability

✓

Higher cash flow generation

✓

Higher IC, liquidity

✓

Higher leverage ratios

✓

IC = interest coverage ratio

④ Security features

Feature	Lower	Higher
---------	-------	--------

Debt

Callability

✓

→ benefits company - call debt at lower rates

Putability

✓

→ benefits bondholder - put debt at higher rates

Convertibility

✓

Equity

Cumulative

✓

Preferred

Inferior cash flow or voting rights

✓

Common

(dual class shares)

* Explain top-down and bottom-up factors that impact the cost of capital

• Estimating cost of debt/

• method depends on:

1/ Type of debt a) publicly-traded → straight debt (no optionality)

- YTM on longest maturity existing debt

or// if shorter-term bonds are more liquid (trade more often), YTM on these may be more reliable

b) non-traded debt

• credit rating approach - use YTM on bonds of other companies with the same credit rating and maturities

- use matrix pricing to estimate YTM

actual cost of debt may differ from company rating based on features of the issue itself

• synthetic credit rating approach - use company fundamentals (IC, leverage) to deduce likely bond rating

- requires a bond rating model

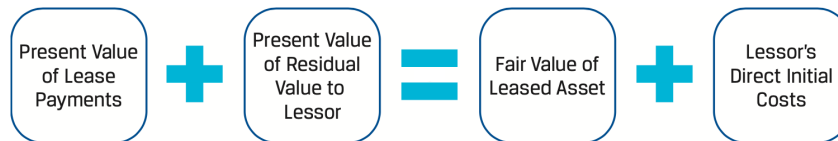
- then use matrix pricing or current credit spread for that rating/maturity (add to r_f)

* Compare methods used to estimate the cost of debt

1/ Type of debt

c) **Bank debt** - quoted interest rate if belief is it reflects current market conditions and the company's current risk profile

d) **Leases** - finance (capital) lease is basically a secured loan
- **RIIL** - rate implicit in the lease may be used but may be unknown (if public company - interest rate on leases usually disclosed)



RIIL - rate that equates both sides

- if **RIIL** unknown - use the incremental borrowing rate (**IBR**)
 - the rate on a secured loan
- if also unknown, may use non-traded debt estimate

* Compare methods used to estimate the cost of debt

2/ **International Considerations** - cost of debt should reflect the currency in which the company's cash flows occur

- add a country risk premium using a country risk rating (**CRR**)
 - considers: economic conditions
 - political risk
 - exchange rate risk
 - security market development/regulation

• **ERP/** $r_e = r_f + (ERP + IRP)$

1/ **historical approach** - mean value of broad-based equity market index return minus government debt - over some historical period

- assumes returns are stationary and markets are relatively efficient

- **decisions/**

1/ **equity index selection** - broad-based, market-value weighted indexes

* Explain historical and forward-looking approaches to estimating an equity risk premium

- decisions/

2/ Time period • largest reliable return series available
but// distant past may not be representative of
the current market environment

- shorter period - increases likelihood of greater noise in
the ERP estimate (unusual time period,
atypical returns)

- longer series preferred - reduces standard error of estimate
(i.e. increases precision)

3/ Selection of the mean type

Mean Type	Advantages	Disadvantages
Arithmetic Mean	• Easy to calculate • Considers all observations in the time series	• Sensitive to extreme values • Overestimates the expected terminal value of wealth
Geometric Mean	• Considers all observations in the time series • Gives outliers less weight • Estimates the expected terminal value of wealth	

→ best for one-period return estimates

- consistent with CAPM

• both are used in practice

* Explain historical and forward-looking approaches to estimating an equity risk premium

- decisions/

4/ Selection of the risk-free proxy

T-Bill - short-term ~ 30-90 days

Treasury - long-term ~ 20 yrs. - preferred

Risk-Free Proxy	Advantages	Disadvantages
Short-term government bill rate	• The rate is an exact estimate of the risk-free rate, assuming no default.	• The rate does not closely match the duration of an infinite-life equity security.
Long-term government bond YTM	• The YTM more closely matches the duration of an infinite-life equity security.	• The YTM is not a completely risk-free return at the time of purchase because of unknown coupon reinvestment rates.

Limitations/ ERPs can vary over time → may display regime changes

Survivorship bias → tends to inflate historical estimates
(index deletions/additions)

• Forward-looking/ • ERP is estimated using current info and expectations about those inputs

- consistent with the notion that the ERP depends strictly on future expectations

* Explain historical and forward-looking approaches to estimating an equity risk premium

- 1/ **Survey-based methods** - sampling of experts about their expectations for the ERP or their CMEs from which the ERP can be inferred
- can be sensitive to recent market returns

2/ **Dividend Discount Method (DDM)**

$$V_0 = \frac{D_1}{r - g}$$

Gordon growth model (constant growth)
- assumes a constant P/E

$$r = \frac{D_1}{V_0} + g$$

income + growth
consensus analyst expectations

$$ERP = E\left(\frac{D_1}{V_0}\right) + E(g) - r_f$$

- can be adjusted for ΔPE and ΔS

- rapidly growing companies - multi-stage growth model (L2 - Equity)
(fast, transition, mature)

* Explain historical and forward-looking approaches to estimating an equity risk premium

3/ **Macroeconomic modelling • Grinold-Kroner:**

$$ERP = [DY + \% \Delta PE + i + g - \% \Delta S] - r_f$$

Dividend yield

change in P/E
(= 0 in the LR)

rGDP growth

change in share count
(= 0 in the LR)

$$\left(\frac{1 + YTM_{Treasury}}{1 + YTM_{TIPS}} \right) - 1 \sim YTM_{Tr.} - YTM_{TIPS}$$

Limitations/

Forward-Looking Approach

Limitation

- | | |
|----------------------|--|
| Surveys | <ul style="list-style-type: none"> • Estimate data can be subject to sampling and response biases, and to behavioral biases such as recency bias (placing more relevance on recent events) and confirmation bias (paying more attention to information that supports one's opinions and ignoring the rest). |
| DDM | <ul style="list-style-type: none"> • Assumes constant P/E. where growth in earnings, dividends, and prices are different from one another; an adjustment is needed to reflect P/E multiple expansion or contraction. |
| Macroeconomic models | <ul style="list-style-type: none"> • Financial and economic models could have modeling errors or behavioral biases in forecasting. |

* Explain historical and forward-looking approaches to estimating an equity risk premium

1/ DDMs $r_e = D_1/V_0 + g$ - company level variables
- requires a publicly-traded company with a stable dividend

or/

$$P_0 = \sum_{t=1}^n \frac{D_t}{(1+r_e)^t} + \frac{P_n}{(1+r_e)^n} \quad \text{- solve for } r_e$$

2/ Bond yield plus risk premium (BYPRP) approach

$r_e = r_d + RP$ r_d - company's long-term debt
RP - equity index ret. - corporate bond index ret.
- for a company with publicly-traded debt

Advantages

- Estimating a company's cost of debt provides a starting point estimate of the return demanded by that company's ~~debt~~ equity investors.

Disadvantages

- Determination of RP is relatively arbitrary.
- Approach requires company to have traded debt.
- If the company has multiple traded debt securities, each with different features, there is no prescription regarding which bond yield to select. Common practice is to use the company's long-term bond YTM.

* Compare methods used to estimate the required return on equity

3/ Risk-based models

CAPM/ $r_e = r_f + \beta(ERP)$

regression estimate

$$(r_e - r_f) = b_0 + b_1(R_m - r_f)$$

what index proxy
market model
what time period

- private or publicly-traded companies
comparable company β - delever/relever

Fama-French models (FFM)/

$$r_e = r_f + b_1ERP + b_2SMB + b_3HML + b_4RMW + b_5CMA$$

size (sm. - lg.) value (high B/P - low B/P) profitability (robust - weak) investment (conservative - aggressive)

- b_1 will differ between CAPM & FFM
- partial reg. coeff. (not by much)
- r_e will differ based on model choice
- r_f tends to be a short-term rate when computing $(R_m - r_f)$

* Compare methods used to estimate the required return on equity

Private Companies/ • prices/returns not available

- tend to be smaller with more owner-centric governance
- less liquid/transparent

- required return will include
 - size premium (SP) *more difficulty securing capital, more uncertain growth prospects*
 - industry risk premium (IP)
 - company-specific risk premium (SCRIP) *e.g. geographic, key person risk*

- illiquidity reflected in a DLOM (discount for lack of marketability) on the price

1/ Expanded CAPM:
$$r_e = r_f + \beta_{\text{peer}}(\text{ERP}) + \text{SP} + \text{IP} + \text{SCRIP}$$

delever, relever (pointing to β_{peer})

no betas (pointing to β_{peer})

if warranted (pointing to SP + IP + SCRIP)

may be proxied by RP for micro-cap companies (typically overstates the RP) (pointing to SP + IP + SCRIP)

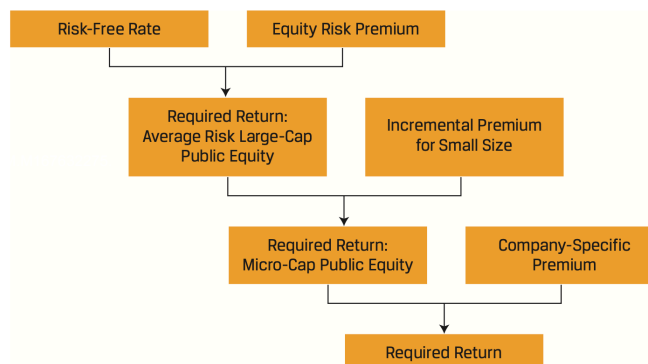
* Compare methods used to estimate the required return on equity

1/ Expanded CAPM: company-specific factors

Qualitative Factors	Quantitative Factors
• The industry in which the business operates • Competitive position within the industry • Management's experience and expertise • Customer and supplier concentration • Geographic concentration of the business • Governance model of the company • Asset nature and type (tangible vs. intangible)	• Financial and operational leverage • Volatility in cash flows and earnings • Earnings predictability • Pricing power

2/ Build-up approach:

- when comparable public companies are unavailable or of questionable comparability



* Compare methods used to estimate the required return on equity

International Considerations/ for EM companies:

$$ERP = ERP_{Dev.} + (\lambda \times \text{Country risk premium})$$

level of exposure of the
company to the local country

Yield_{EM} in dom. curr.

- Yield_{DM}

does not capture
differences in legal and
market environments

Damodaran:

$$CRP = \text{Sov. yield sp.} \times \frac{\sigma_{equity}}{\sigma_{bond}} > \text{local markets}$$

→ **Extended CAPM** - for companies operating internationally

- **GCAPM** - global CAPM - use of a single global market factor
- assumes no significant risk differences across countries

* Compare methods used to estimate the required return on equity

• **ICAPM** - International CAPM

$$E(r_e) = r_f + \beta_G(E(R_{gm}) - r_f) + \beta_e(E(R_e) - r_f)$$

observable (domestic)

stock return (observable)

global index

MSCI All Country World Index (available series)

country wealth weight

domestic

returns to a for. curr. vs. the domestic curr.

lower values
associated with
companies less connected
to the global economy
(regression)

- assumption - all economies are fully integrated into the global economy

- effects from holding an asset in a different country
- effects on business profits and returns (changes in a currency can affect profitability (imports/exports))

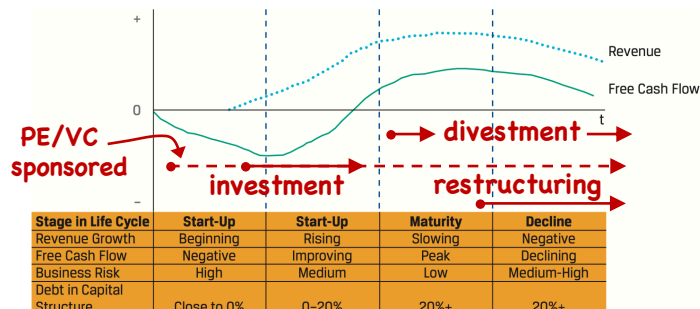
* Compare methods used to estimate the required return on equity

Corporate Restructuring

- a. explain types of corporate restructurings and issuers' motivations for pursuing them
- b. explain the initial evaluation of a corporate restructuring
- c. demonstrate valuation methods for, and interpret valuations of, companies involved in corporate restructurings
- d. demonstrate how corporate restructurings affect an issuer's EPS, net debt to EBITDA ratio, and weighted average cost of capital
- e. evaluate corporate investment actions, including equity investments, joint ventures, and acquisitions
- f. evaluate corporate divestment actions, including sales and spin offs
- g. evaluate cost and balance sheet restructurings

Corporate Restructuring

Page 1/



Investment - actions that increase the size of the company or scope of operations
(focus is on external investment, non-organic growth i.e. acquisition)

Divestment - actions that reduce a company's size or scope by shedding slower growing or higher risk operations e.g. sale of a division

Restructuring - changes that do not alter the size or scope but improve its cost and financing structure with the intent to increase growth, improve profitability, or reduce risk
e.g. improve productivity

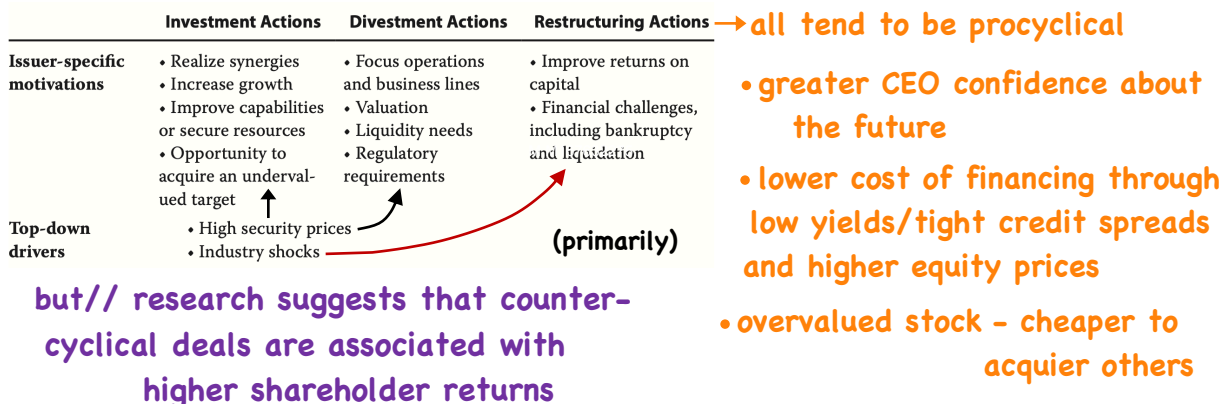
* Explain types of corporate restructurings and issuers' motivations for pursuing them

Page 2/

• large corporate issuers → portfolio of many diverse lines of business at different stages of the life cycle and operate in different competitive environments

may be synergies
(acquire)

may be costs of scope and inefficiencies
(divest, restructure)



* Explain types of corporate restructurings and issuers' motivations for pursuing them

• Types of Restructurings/

- Investment actions:

• motivations

1/ creating cost or revenue synergies
typically through redundancies and economies of scale and economies of scope (e.g. cross-selling)

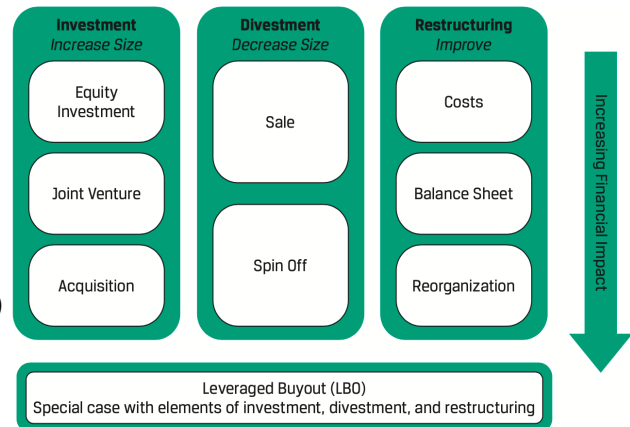
2/ increasing growth

3/ improving capabilities

4/ access to resources (when scarce)

(vertical integration) → becomes a divestment target once resource is abundant

A/ Equity Investment - buying a stake in another company → influence but not control
- may be strategic, pre-acquisition, or financial
(> 20%, < 50%)



* Explain types of corporate restructurings and issuers' motivations for pursuing them

B/ Joint Venture - 2 or more companies form a new company to achieve some business objective

- usually to enter new markets or share a common, non-competitive advantage

e.g./ GM + LG - build battery plants in U.S.

C/ Acquisition - one company purchases most or all of another company

- control position (> 50%)

- target becomes a subsidiary of the acquirer

- financial results of target consolidated at parent level

- divestment actions/ motivations

• increase valuations - unlock value - a pure play bus. will trade at a higher multiple

• liquidity

• regulatory requirement

typically the result of external circumstances - over leveraged or anti-competitive

* Explain types of corporate restructurings and issuers' motivations for pursuing them

- sell or spin off - investors in the company become shareholders of the second company (for cash)
(e.g. HR.UN - spins out certain mall properties into PMZ.UN)
- conglomerate discount - a large issuer that trades at less than the sum of the parts
- generally due to diseconomies of scale or scope
- A/ Sale (divestiture) - control of assets/segment/division transfers to acquirer → parent gets cash
- B/ Spin-off - a distinct part of the business separated to form a new independent company → no cash for parent
- shares of Comp. A now become shares of Comp. A and B

* Explain types of corporate restructurings and issuers' motivations for pursuing them

- restructuring/ motivations:
 - opportunistic improvement
 - forced improvement → when $ROE < rrr$
- alter business model
 - trim cost structure
 - modify composition of BS

} improve ROC

pro-active
- dropping demand
 - competitive pressure
 - over-capacity

reactive
- A/ Cost restructuring - reduce costs by improving operational efficiency
- improve margins
- a) outsourcing - typically standardizable business processes
e.g. IT, delivery, call centers
 - reduces headcount, costs, oversight time
 - reduces asset size - office/warehouse space, equipment

* Explain types of corporate restructurings and issuers' motivations for pursuing them

b) offshoring - relocating operations to another country
- usually low labour-cost locations

B/ Balance sheet restructuring - alter the composition of the BS

shift asset composition

e.g./ sale - lease back - sell an asset
for cash and enter into a lease
- higher cost than owning the asset
but// cash can be put to higher return use

change capital structure

e.g./ dividend recapitalization
- issue debt to pay dividend or
buy back shares
- reduces wacc but raises leverage

C/ Reorganization - court supervised restructuring for insolvent companies
(bankruptcy court)

- business operations continue during the process
- once agreement with creditors is reached, gets court approval, exits bankruptcy, continues operations with lighter debt
- if that process fails - enters liquidation

* Explain types of corporate restructurings and issuers' motivations for pursuing them

- leveraged buyouts - special case of a corporate restructuring
 - acquirer uses a significant amount of debt to finance an acquisition → divests some parts, pays down debt over a 3-5 year period with the goal of an IPO or sale
 - if the target is public, called a 'take-private' transaction
 - typically led by a PE firm (buyout fund)

Initial Evaluation/ what and why → found in securities filings, press releases, conference call transcripts

- is it material? size: larger the restructuring relative to company size, the more likely it is to affect future cash flows
e.g./ transaction value > 10% of EV
- fit: less it fits with existing operations, the more material it may be strategically

* Explain the initial evaluation of a corporate restructuring

timing: timing between announcement and completion of transaction

- approvals may be required as well (shareholders, creditors, regulators)

Preliminary Valuation/ for restructurings that are material and involve transactions

- methods:

1/ **Comparable Company analysis** - use the valuation multiple of similar listed companies to value the target

- need a set of comparable companies - may be in same industry or similar industries with similar financial characteristics
- then calculate valuation multiples based on current market prices

e.g. EV/EBITDA EV/Sales P/E EV/FCFF

↓
less sensitive to
capital structure differences

→ may even be industry specific
e.g. EV/FFO for real estate

* Demonstrate valuation methods for, and interpret valuations of, companies involved in corporate restructurings

- apply mean/median multiple to target

- typically for spin-offs (acquisitions involve a premium) **ex. #7**

Advantages:

- provides a reasonable approximation of value
- most of the required data is available
- market derived estimates of value (rather than assumptions)

Disadvantages

- comparable list of companies may be difficult to find or may not exist
- sensitive to market mispricing
- fair trading price only, not a fair takeover price

DCF

2/ **Comparable transaction analysis** - use valuation multiples from historical acquisitions of similar targets

- multiples would already include takeover premiums

Ex. #8

* Demonstrate valuation methods for, and interpret valuations of, companies involved in corporate restructurings

Advantages: • data come from actual transaction prices
(not assumptions or estimates)

• no takeover premium needs to be estimated

Disadvantages: • may be no/few comparable transactions

• historical multiples may not reflect current
industry stage/macro environment

• past acquirers may have over or under paid

3/ Premium paid analysis/

$$\text{PRM} = \frac{\text{DP} - \text{SP}}{\text{SP}}$$

deal price

unaffected stock price

takeover premium

- from historical transactions

- reflects price of control

must exclude any
pre-announcement increase

- use P one week (or longer)
before announcement

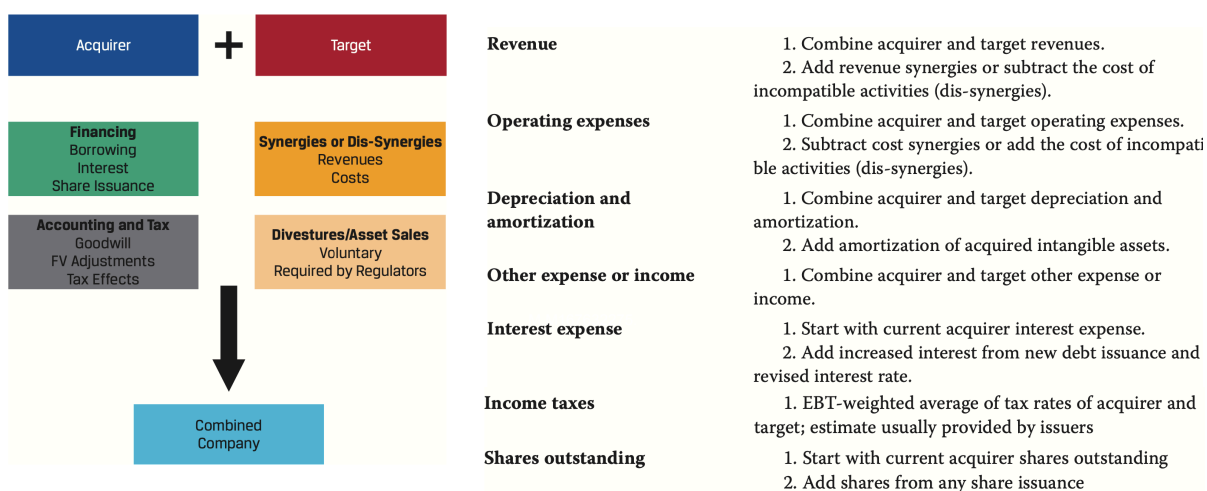
- apply avg. premium to target's price

* Demonstrate valuation methods for, and interpret valuations of, companies involved in corporate restructurings

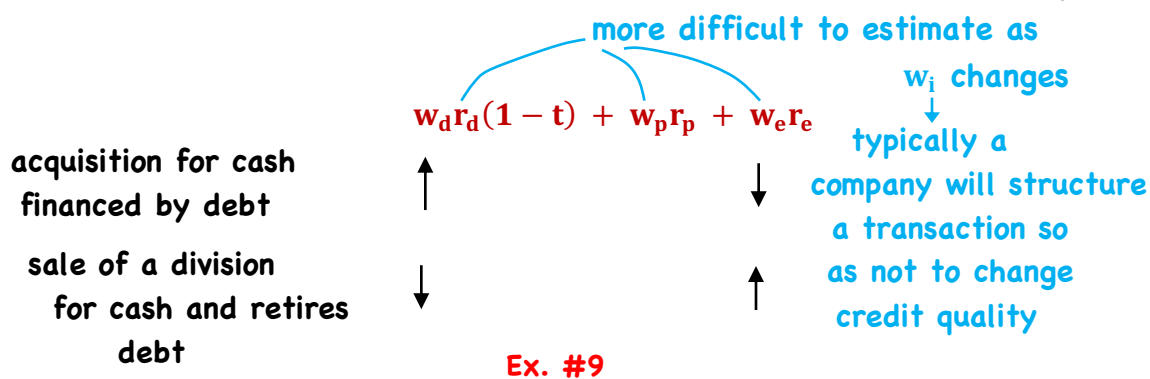
- next step → estimate pro-forma financial statements

include the effect of the restructuring

- process depends on the type of restructuring



* Demonstrate how corporate restructurings affect an issuer's EPS, net debt to EBITDA ratio, and weighted average cost of capital



* Evaluate corporate investment actions, including equity investments, joint ventures, and acquisitions

Ex. #10

Ex. #11

Ex. #12

* Evaluate corporate divestment actions, including sales and spin offs **Ex. #13**

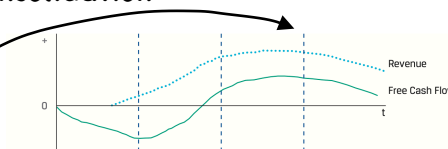
* Evaluate cost and balance sheet restructurings **Ex. #14**

* Demonstrate how corporate restructurings affect an issuer's EPS, net debt to EBITDA ratio, and weighted average cost of capital

Ex. #10/ 30% for 1.2B (4B EV) - equity method
- no consolidation

1/ acquisition, JV (ex. #5)

2/ Dilmun - exposure to a growth category
Spina - access to growth channels



3/ $EV/sales \rightarrow 4B/600 = 6.667$

Spina 20X3 Rev.

(explain how, not why)

4/ pay more — higher g - 225k locations from 75k + mktg.
lower r material inside cig. packs
↳ access to growth channels lowers risk of growth

5/ Current $D/EBITDA = 13894/10140 = 1.3702$

diluted EPS = $7256/1840 = 3.94$

Exib. #22

Ex. #10/ 5/ New $D/EBITDA = \frac{13894 + 1200}{10140 - 135} = \frac{15,094}{10,005} = 1.50864$

(30% of 450)
loss

EBIT	9890		(ex. #22 20X4E)
-	135	←	
-	10	←	(added Amm.)
	9755		
I =	(705)		
6% of 1.2B	(72)		
	8968		
T(21%)	1883		
NI →	7085	→	$7085/1840 = 3.8504$

$\text{Tax rate} = 1929/(9890 - 705) = 21\%$

Ex. #11/

1/ Opone - not profitable without it

Hapalla - access to foreign market, partner to share risk

2/ outsized Div. from JV (in excess of NI in 20X7)

- boost to CF > boost to NI

- not expected to continue → but still has the cash

3/ $EV = MV_e + MV_d - C = 147,359$ (example is wrong)

180B 0 32461

(45B for 25%) not equity

$P/E = 180B/15.267 = 11.79$ vs. comparables 12x

(not EV/E)

Ex. #11/ (mill.)

4/	Rev.	3910
	COGS	(3793)
	SG&A	(1450)
Income from associates		4689
Gain on sale		32,000
I		(95)
EBT		35261
T(10%)		3526
		31735

ex. #28 20X8E

→ 50% of 20X8E

26B for 50% under JV

- 13B for 25%, getting 45B

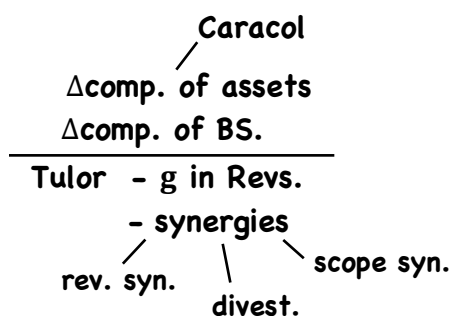
5/ FRA Question

Happalla: equity method to consolidation

- no 'Investment in Associates'

- combined A & L

Ex. #12/ 1/ motivations



2/ median comparable = 8

Caracol 3B/183 - ex. #33 20X2

2BC = 16.39

1B sh.

- control premium + PV(synergies)

- example incorrect in adding synergies in YR3 to current EBITDA

3/ 20X3

Rev.	27158
COGS	(20819)
Op.Ex.	(4345)
+ Synergies	42
EBITDA	2036
D/A - T	(652)
D/A - C	(109)
new	(200)

3800 x .005

1B x .065

I

T (20%)

NI

Ex. 32	
+ Ex. 33	
for 20X3E	
D/EBITDA	
= (6169 + 1000)	
2036	
= 3.5211	

20X3E

4800 - 1B cash

- new debt @ 6.5%

ex. 32 20X3E

502 / (648 + 80) = .6895

old new sh. (vs. 96)

Ex. #13/ 1/ EV = 1.437B

- unallocated all to consulting? - has to go somewhere

could be allocated by Sales, - can't go to BE - EBITDA < 0
but almost all would be consulting anyway ∴ EV/Sales used

BE Sales 75 (Ex. #37 LTM)

Peer median x 12/900M

Consulting EBITDA 84 (Ex. #37 LTM)

unallocated 4
80

Peer median x 9
720M

$$\begin{array}{r} 900 + 720 = 1.620B \\ - EV \quad 1.437B \\ \hline 183M \end{array}$$

$$2/ g = \left(\frac{75}{55} \right) - 1 = 36.4\% \quad (\text{ex. \#37})$$

closest comp. w/ g = 35% → E (ex. #42)

15x

$$\begin{array}{r} (75 \times 15) + (80 \times 9) = 1.845B \\ - 1.437B \\ \hline 408M \end{array}$$

Ex. #13/ 3/ a)

Comparable transaction	Cash paid	Value of stock issued	Net debt (cash) assumed	Target EBITDA (LTM)
Comparable 1	791	0	118	101
Comparable 2	(1,174	+ 0	+ 434)	÷ 134
Comparable 3	(578	+ 84	+ (35))	÷ 87
Comparable 4	(1,310	+ 378	+ 832)	÷ 180

$$\begin{array}{r} 791 + 0 + 118 = 909 \\ \hline 101 = 9 \\ = 12 \\ = 7.2 \\ = 14 \end{array}$$

Med. = 10.5
Mean = 10.55

$$\text{Consulting bid } 800 \div 84 = 9.523 \text{ EBITDA}$$

low

b) implied valuation of the consulting segment
in 1.437B using (75M × 15) for BE

$$1.437B - 1.125B = 312M \quad \text{vs. } 800M \text{ bid}$$

Ex. #13/ 4/

BE	EBITDA	(5)	
unallocated 4		(3)	
- 1 / 3		(8)	
D/A	(20)	(8)	
- 12 /		(16)	
I		(9)	
NI	→	(25)	→

original EPS
 $35M / 1.454B = 2.41 \text{ ¢}$

$-25M / (1.454B - .2B) = -1.99 \text{ ¢}$

5/ Consulting $EV / EBITDA = 13$ $\therefore EV = 13 \times 80 = 1.040B$
vs. 800M

but// spin-off → non-cash transaction
 - market may not support that valuation

Ex. #14/ 1/

A	30%	
B	24.8%	mean 22.26%
C	17.4%	
D	20.3%	→ med.
E	18.8%	

Cyrene 14%
 ↓
 valuation reflects this

2/ med. = 20.3%
 $- 14\% / 6.3\% \times 18,990 = 1,196.37$ in cost reductions needed
 $- 2659 / = 16,331$ Total costs

$1196.37 / 16,331 = 7.325\%$

3/ Rev. 19559.70 $(18,990 \times 1.03)$
EBIT (ex. rest.)
 $14\% + \frac{1}{4}(6.3\%)$ 3046.42
Restructuring 1250.00
 1796.42 → **9.18% of Rev.**

Ex. #14/

4/ growth costs money → cost reduction may reduce growth

pacing
 $\swarrow$ too slow
 $\searrow$ too fast

political - ?

5/ HG 9% g = 7%

BPC 24% g = 3%

F 11% g = 2%

can these grow faster

what are industry growth rates

what are peer margins

Rev. w/° HG
13,483

EBIT w/° HG
2163

16% margin (+200 bps)

• all unallocated costs?

• value of HG
 $\swarrow$ sale
 $\searrow$ spin-off
 $\searrow$ left in
 ?

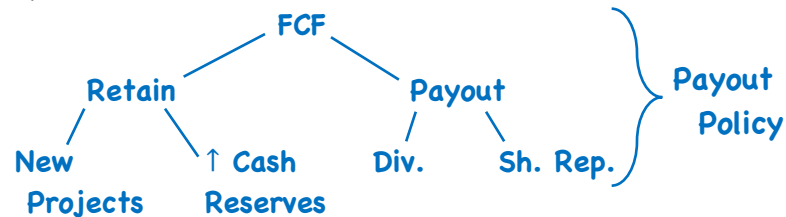
• benefits/costs to BPC/F?

REVIEW

Dividends/Share Repurchases

Review - 1

- does payout policy affect firm value?



1) Dividend Irrelevance Theory (MM,61)

- assume perfect capital markets - no taxes

⇒ div. policy has no effect on CoC or shareholder wealth

(div. & sh. repurch. are interchangeable)

⇒ Investors can create their own dividends

trans. costs
agency costs
+ info symmetry

2) Dividend Preference Theory - stock risk declines as div. ↑

- dividends are a sure thing

(bird in the hand argument)

∴ $r_e \downarrow$, $wacc \downarrow$

- also reduce agency costs

3) Tax Effect Theory

$T_D > T_{CG}$ - investors prefer low payout ratios (willing to pay more)

$T_D < T_{CG}$ - investors prefer cash dividends

Review - 2

⇒ Dividend Signals/ dividend initiation/increases - signal that mgmt.

- may be some info. in. Div. expects higher future earnings

Policy changes/ - dividend omissions/decreases/failures to increase

- convey negative information

⇒ Agency Costs/ - Jensen's FCF hypothesis - high payout ratio

constrains mgmt. from overinvestment

But// - high payout ratio increases agency costs to creditors (may also constrain available cash leading to underinvestment)

⇒ Factors Affecting Payout Policy/

1) Investment opportunities - more = lower payout

- RE are a cheaper source of funding

⇒ Factors Affecting Payout Policy/

2) expected volatility of future earnings – more volatile, more risk that dividend increase may not be met by future earnings

3) financial flexibility – share repurchases rather than initiate a dividend

4) tax considerations

⇒ taxation methods (pg. 16-17)

double taxation / corporate then indiv.
dividend imputation – incremental tax only
split rate – 2 rates on earnings

5) Floatation Costs – try to avoid level of dividends that would create the need to raise new equity

6) Contractual/Legal Restrictions

- some countries – co. must pay a div.
- others restrict payment (impairment of capital rule)

Taxation Systems

1) Double Taxation System/

	39.6%	15.0%
Net income before taxes	US\$100	US\$100
Corporate tax rate	35%	35%
Net income after tax	US\$65	US\$65
Dividend assuming 100% payout	US\$65	US\$65
Shareholder tax on dividend	US\$25.74	US\$9.75
Net dividend to shareholder	US\$39.26	US\$55.25
Double tax rate on dividend distributions*	60.7%	44.8%

- different personal tax rates

Page 16

LOS e
- calculate
- interpret

2) Dividend imputation tax system/

	Marginal Shareholder Tax Rate	
	15%	47%
Pretax income	A\$100	A\$100
Taxes at 30% corporate tax rate	30	30
Net income after tax	70	70
Dividend assuming 100% payout	70	70
Shareholder tax on pretax income	15	47
Less tax credit for corporate payment	- 30	- 30
Tax due from shareholder	(15)	17
Effective tax rate on dividend	15/100 = 15%	47/100 = 47%

- one higher, one lower than corporate rate

- tax credits can be used to offset taxes owed on other income

3/ Split-rate tax system

Pretax earnings	€200
Pretax earnings retained	100
35% tax on retained earnings	35
Pretax earnings allocated to dividends	100
20% tax on earnings allocated to dividends	20
Dividends distributed	80
Shareholder tax rate	35%
After tax dividend to shareholder	$[(1 - 0.35) \times 80] = 52$
Effective tax rate on dividend	$[20\% + (80 \times 0.35)\%] = 48\%^*$

loss of \$48
= 48% tax

Dividends/Share Repurchases

⇒ Dividend Policy/

1) Stable Dividend Policy/ - most common

- dividends based on long-term forecast of sustainable earnings

$$E\left(\frac{\text{increase in Div.}}{\text{Div.}}\right) = \left[\frac{E(\text{earnings}) \times \text{target DPR} - \text{Previous Div.}}{\text{DPR}} \right] \times \frac{1}{\text{time frame}}$$

moving towards a target DPR

2) Constant Dividend Payout Ratio ← DPR vs. Div.

- div. fluctuate w/ earnings

⇒ Cash vs. Repurchase/

1) Potential Tax Advantage

T_D lower - divs.

T_{CG} lower - cap. gains

2) Share price support

shares undervalued?

few reinvestment options?

⇒ **Cash vs. Repurchase/**

- 3) added managerial flexibility – repurchase not contractual
- 4) Offsetting dilution from employee stock options
- 5) Increasing financial leverage – decrease equity
(modifies capital structure) – increases leverage

– Sh. repurchase an alternative to a special dividend

⇒ Catering Theory – dividend policies change over time to
changing investor tastes

⇒ Dividend Coverage Ratios/ $DPR = \frac{Div.}{NI}$ $DCR = \frac{NI}{Div.}$ $\left. \begin{array}{l} \text{higher DPR, lower} \\ \text{DCR, higher risk} \\ \text{of Div. cut} \end{array} \right\}$

$$FCFE \text{ Cov. Ratio} = \frac{FCFE}{Div. + Repurchases}$$

⇒ unsustainable Payouts/

$DPR > 100\%$, $DCR < 1$, $\frac{FCFE}{(Div. + Repurch.)} < 1$, borrowing to
pay Divs.